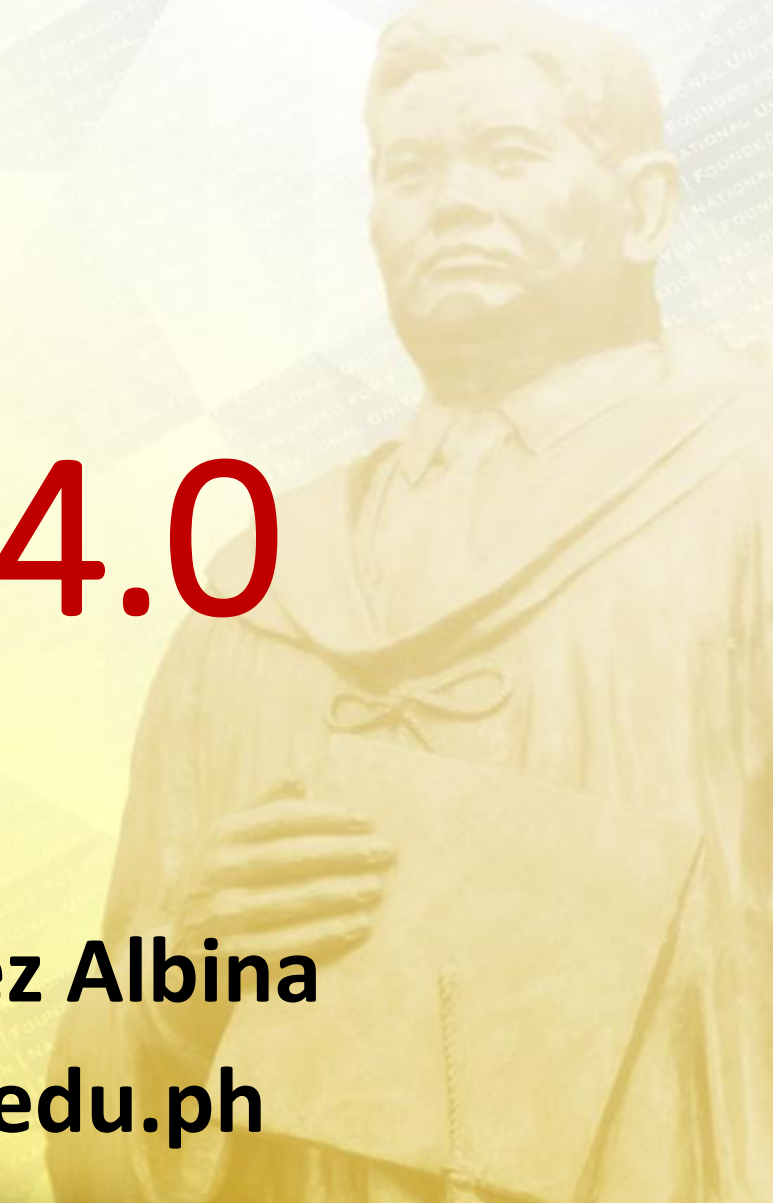
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Week 5 to 6: GEITE01X
Living in the IT Era

Industry 4.0

Professor: Dr. Earl Marquez Albina

Email: ealbina@nu-moa.edu.ph



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Help us to be respectful and kind to one another,
fostering an environment of cooperation and growth.

Agenda for Today:

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1. Opening Prayer
2. Greetings and Checking of Attendance
3. Lesson Recall (Short recitation in random)
4. Validation and Recording of Midterm Quiz No.3
5. Project Presentation
6. Q and A / Quiz and Attendance Concerns
7. New Lesson – Week 6 and 7
8. Closing Prayer

Midterm Exam:

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Coverage Week 1 to 6 PPT

Lecture and Presentation

Midterm Exam Week:

August 24 – 29, 2026

LEARNING OUTCOMES

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At the end of this lesson, you should be able to:

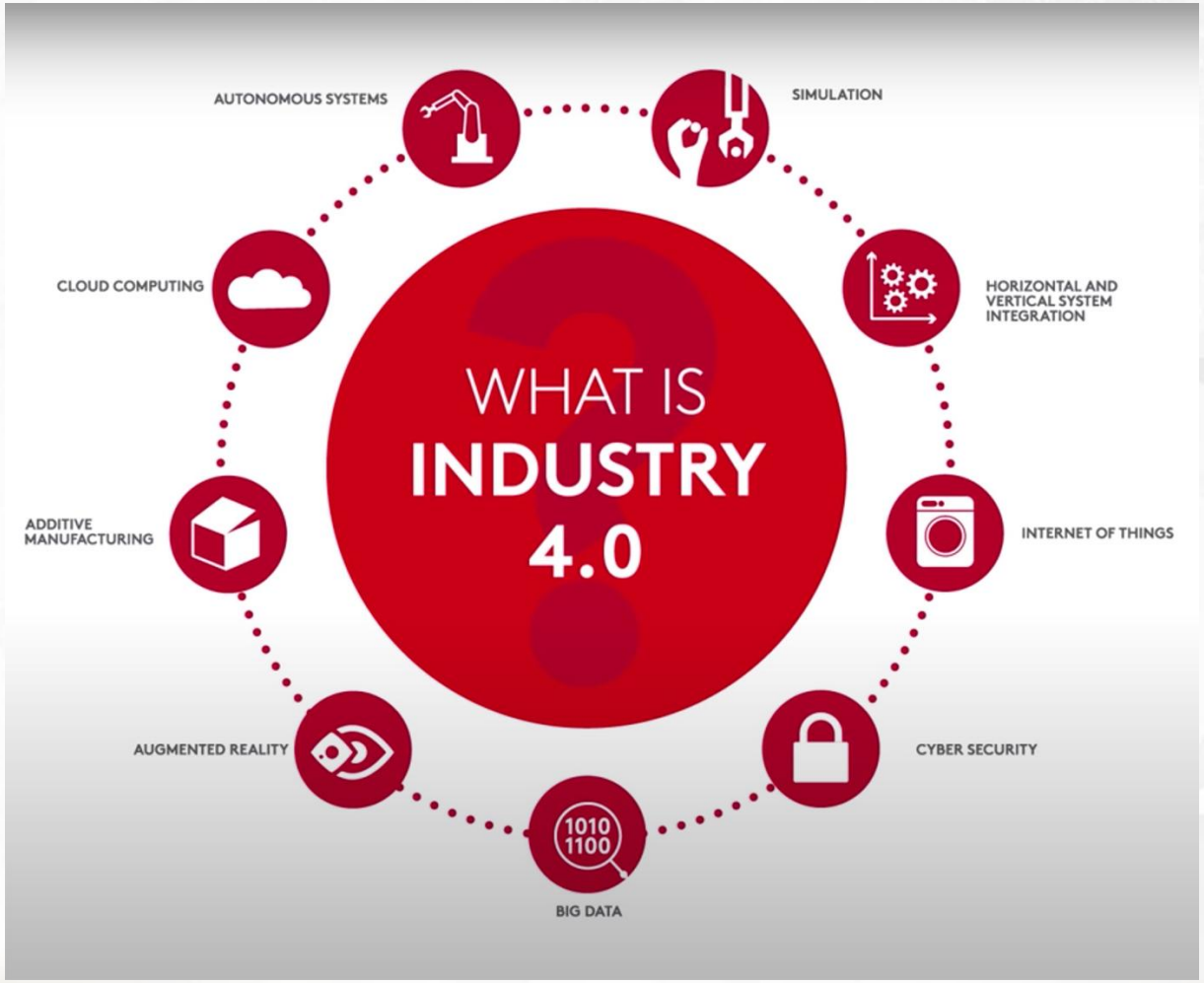
- Be familiar with the phases of the Industrial Revolution
- Understand key technologies related to Industry 4.0

ORIGINS AND HISTORY

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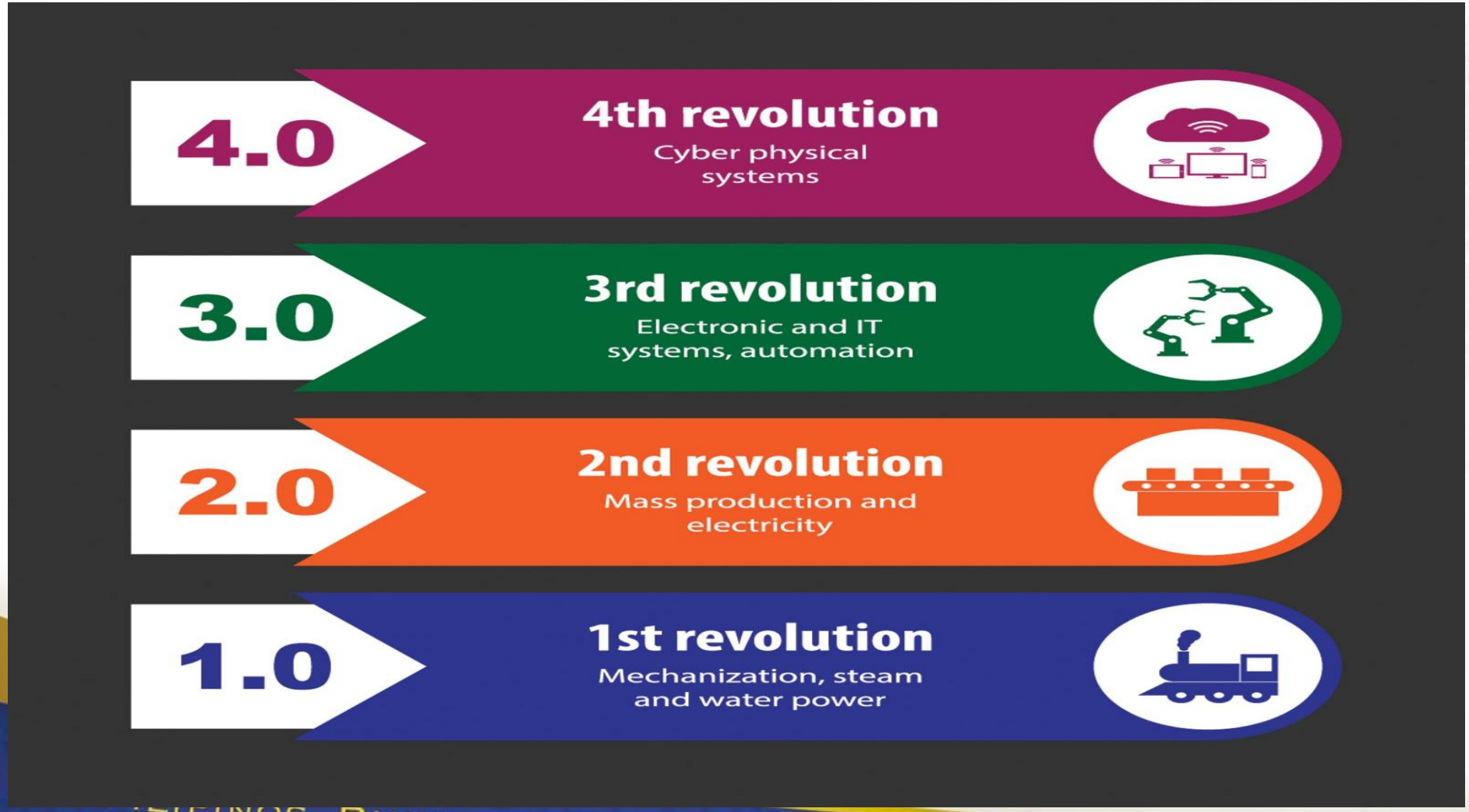
What is Industry 4.0?

- Industry 4.0 is the **digital transformation of industry and manufacturing**.
- It enables **real-time decision-making**, higher productivity, and greater flexibility.
- Companies use advanced technologies to improve how products are **made, improved, and distributed**.
- Key technologies include:
  - 🌐 Internet of Things (IoT)
  - ☁️ Cloud Computing
  - 📊 Big Data & Analytics
  - 🧠 Artificial Intelligence (AI)
  - 🤖 Machine Learning
- These technologies help create **smart factories and automated operations**.



A SHORT HISTORY OF THE FOURTH INDUSTRIAL REVOLUTION

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FIRST INDUSTRIAL REVOLUTION









- **First Industrial Revolution** started in **Britain during the 18th century** and later spread worldwide.
- The term **"Industrial Revolution"** was popularized by **Arnold Toynbee** to describe Britain's economic growth from **1760 to 1840**.
- It brought major **economic and industrial changes** in Europe between **1780 and 1849**

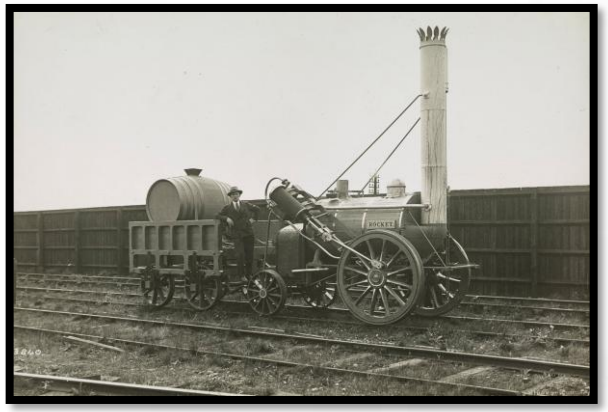
- **Artisan workshops** lost control of production as **new technologies and machines** were introduced.
- Production became **faster, more efficient, and machine-based**, transforming how goods were made

CHARACTERISTICS OF THE 1ST IR

- **Manual work** is replaced by **industrial factory**
- The **railroad** was invented, and it was the revolution in means of transport.
- The **steamboat** was created that allowed the **transportation** of larger ships at higher speed.
- **Roads** and **canals** were built to improve transportation.
- There were great **technological changes** that were the basis of industrial production.
- There was **private investment** and **massive production** of goods.
- There was **low cost** in **production** thanks to machines.
- Low cost of **labor**.
- The new **imperialism** arose, which was based on economic conquest through the sale of industrial products.
- Creation of the **mechanized factory**

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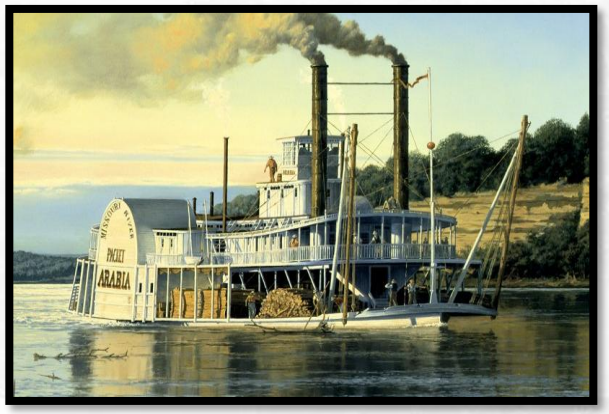
INVENTIONS DURING THE 1ST IR



STEAM ENGINE



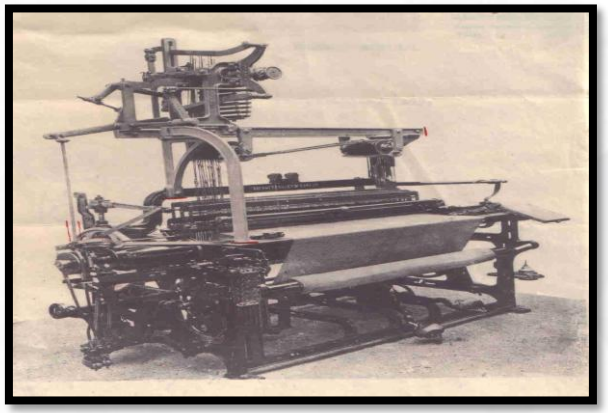
STEAM LOCOMOTIVE



STEAM BOAT

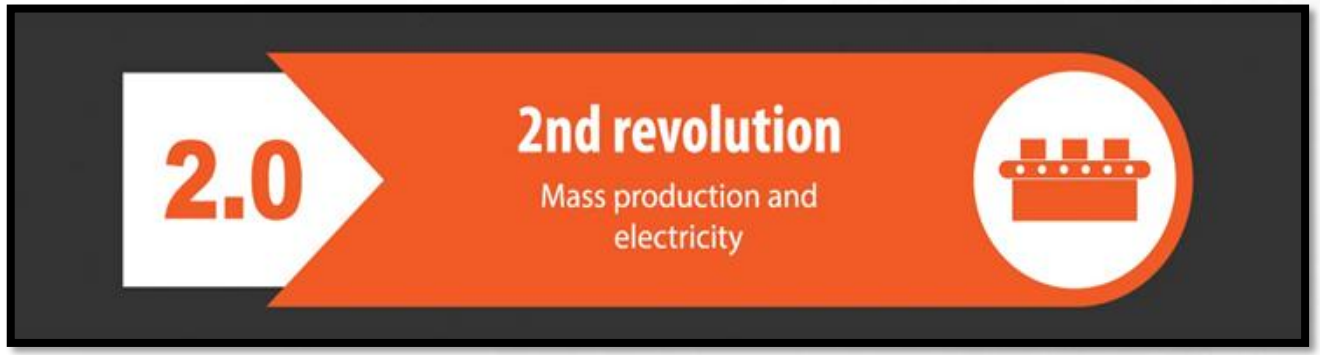


HARGREAVES MACHINE



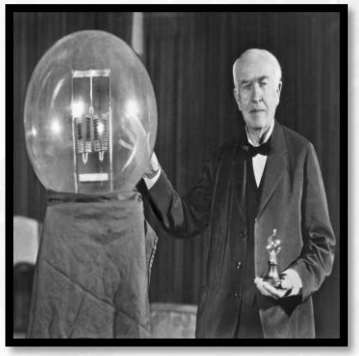
HYDRAULIC LOOM

SECOND INDUSTRIAL REVOLUTION









- **Second Industrial Revolution** began in **England** and spread to **Europe, the USA, and Japan**.
- It lasted from **1870 to 1914**, ending around the start of **World War I**.
- Britain gradually lost its industrial leadership to other major powers.
- Major advances were made in the **steel, oil, chemical, and electrical industries**.
- New technologies led to the development of **automobiles** and **airplanes**.
- It was a period of rapid **industrial growth, innovation, and technological progress** worldwide.

CHARACTERISTICS OF THE 2ND IR

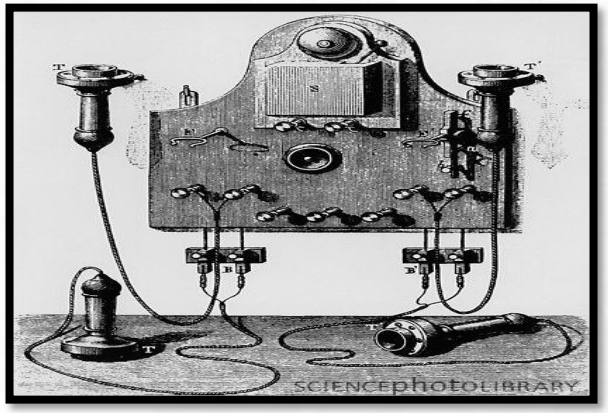
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- Iron was replaced by steel in industry.
- Steam was replaced by electricity and petroleum products were used as an important source of energy.
- Automatic machinery was introduced to manage and operate other machines.
- There were important changes in **transportation** and communications.
- Application of science to industry.
- New forms of capitalist organization were created: imperialism, mechanization and big industry.
- Electricity was discovered.
- Existing machines were automated.
- The airplane was invented.
- Darwin's theory of the evolution of species emerged.

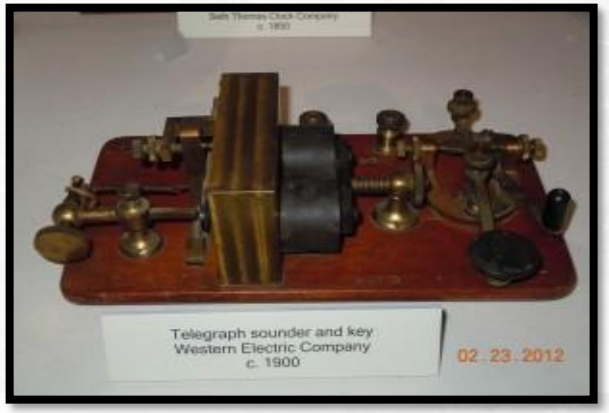
INVENTIONS DURING THE 2ND IR



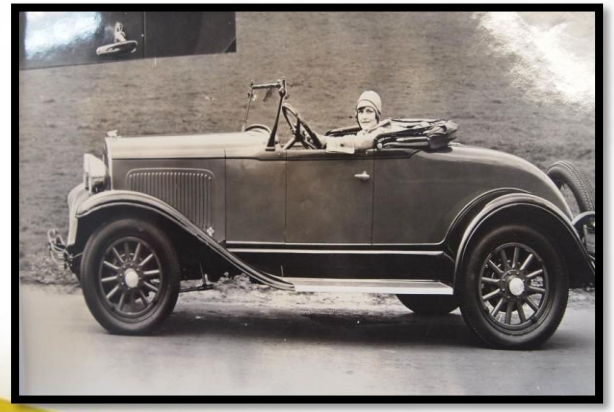
AIRPLANE



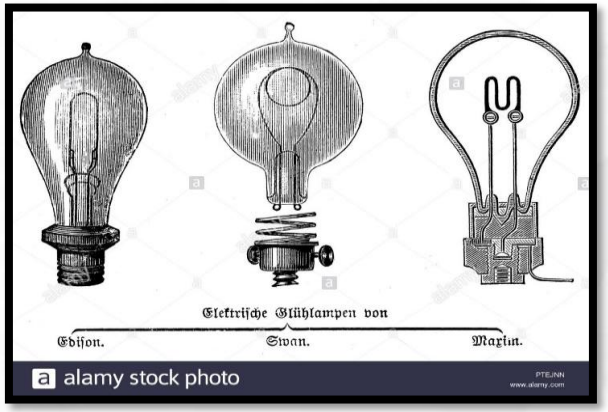
TELEPHONE



TELEGRAPH



AUTOMOBILES



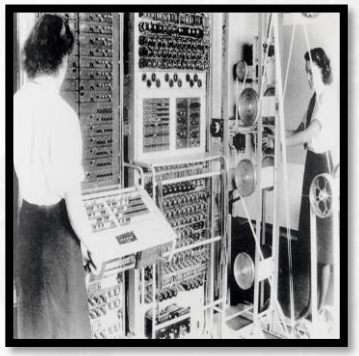
LIGHT BULB

THIRD INDUSTRIAL REVOLUTION









- **Third Industrial Revolution (Digital Revolution)** shifted society from **mechanical and analog technology to digital technology**.
- Began in the late **20th century** with the widespread use of **computers** and the **Internet**.
- Improved **information and communication technologies (ICT)** significantly.
- Connected people worldwide through **instant communication** and access to information.
- Led to technologies such as **smartphones, smart homes, and modern computers**.
- Changed the way people **live, work, communicate, and learn**.

CHARACTERISTICS OF THE 3RD IR

- Information Technology
- Search for Sustainable Energy

INVENTIONS DURING THE 3RD IR



COMPUTERS



INTERNET



SMARTPHONES



DIGITAL TV



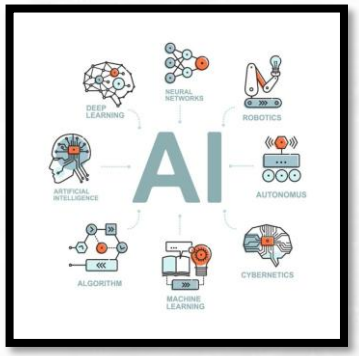
DIGITAL PHOTOGRAPHY



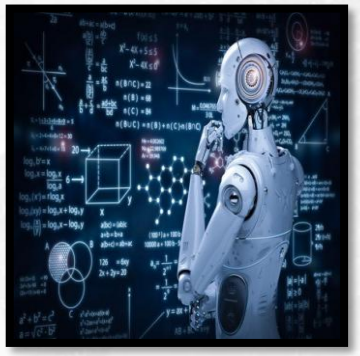
DIGITAL PRINTING

FOURTH INDUSTRIAL REVOLUTION



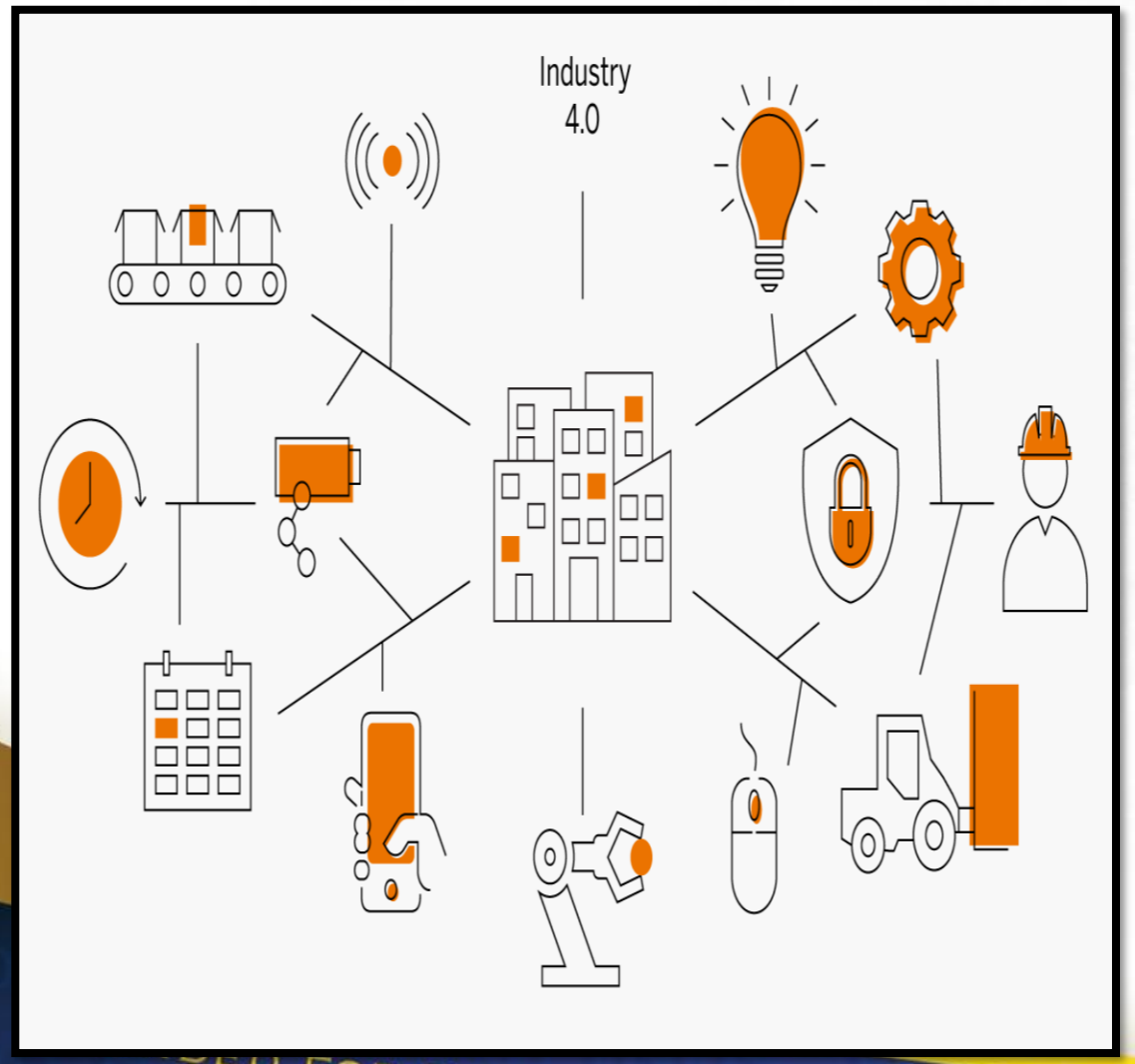






- **Industry 4.0** is the **Fourth Industrial Revolution**.
- Combines **advanced manufacturing** with the **Internet of Things (IoT)**.
- Machines and systems are **connected and communicate** with each other.
- Uses data to **analyze information and make smart decisions**.
- Creates **intelligent, automated, and efficient production systems**.

INDUSTRY 4.0



- **The Fourth Industrial Revolution (Industry 4.0)** focuses on **smart technology and automation**.
- Powered by the **Industrial Internet of Things (IIoT)** and **cyber-physical systems**.
- Uses **smart machines, robots, and vehicles** that can monitor and control processes.
- Makes **factories, warehouses, and logistics systems smarter and more efficient**.
- Connects supply chains with business systems like **ERP** for better visibility and control.
- Plays a key role in **digital transformation** and modern industry.

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Industry 4.0

Week 5 – 6 – GEITE01X – Living in the IT Era

PART 2

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INDUSTRY 4.0 TECHNOLOGIES ARE BUILT ON NINE (9) TECHNOLOGIES

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Industry 4.0 Technologies is built on Nine Technologies

1 Advanced Robotics



Intelligent robots that automate tasks, improve precision and enhance productivity.

2 Industrial Internet of Things (IIoT)



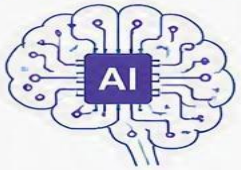
Connected devices that collect, share and analyze data in real time.

3 Big Data and Analytics



Analyzing large volumes of data to drive better decisions and predictions.

4 Artificial Intelligence (AI)



AI enables machines to learn, reason and make decisions like humans.

5 Cloud Computing



Stores and processes data in the cloud for scalability, flexibility and efficiency.

6 Additive Manufacturing (3D Printing)



Creates complex parts and products layer by layer with less waste.

7 Augmented Reality (AR)



Overlays digital information on the real world to assist and improve work.

8 Cybersecurity



Protects systems, networks and data from cyber threats and attacks.

9 System Integration



Integrates machines, systems and software to work together seamlessly.



These nine technologies work together to create smarter, connected, efficient and flexible industries.

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1. ADVANCED ROBOTICS





- Perform tasks with **little or no human intervention**.
- Use **AI, sensors, cameras, and machine vision** to make decisions.
- Can **analyze surroundings and respond automatically**.
- Improve **speed, accuracy, and productivity** in manufacturing.

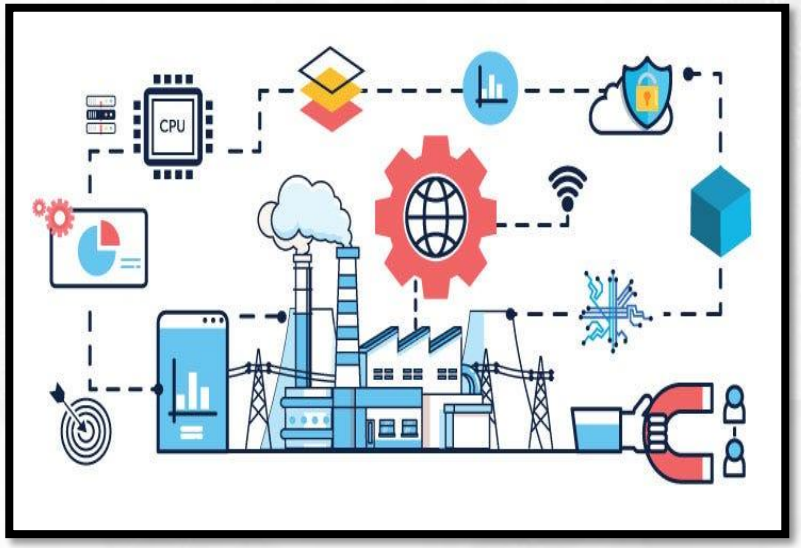
Examples:

- 🤖 **Tesla** - robotic vehicle assembly lines.
- 📦 **Amazon** - warehouse robots that move products.
- 🚁 **Inventory drones** - scan and track warehouse stock.
- 🚚 **Autonomous Mobile Robots (AMRs)** - transport materials in factories.

2. INDUSTRIAL INTERNET OF THINGS (IIoT)

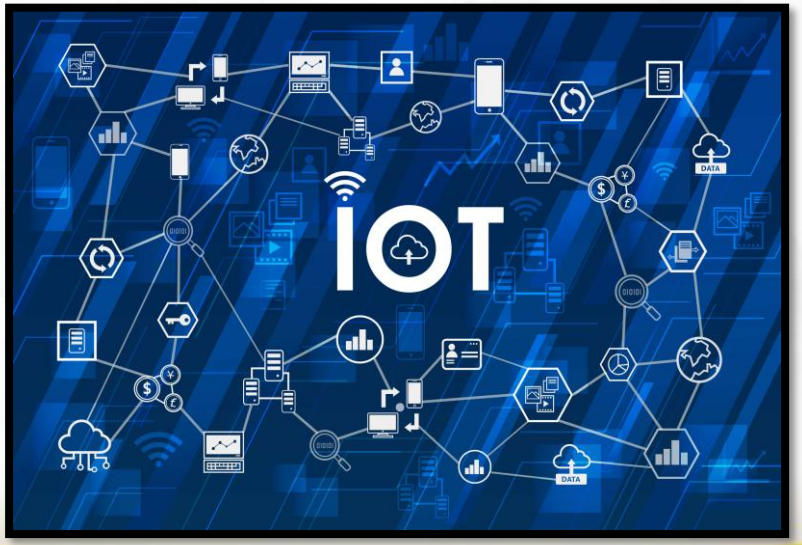
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- Connects **machines, devices, and equipment** through the internet.
- Uses **sensors and RFID tags** to collect real-time data.
- Helps improve **production, inventory tracking, and maintenance**.
- Enables smarter and more efficient operations.



Examples:

- 🏭 **GE** – monitors industrial equipment in real time.
- 🚗 **BMW** – uses connected sensors in smart factories.
- 📦 **Amazon** – tracks inventory and warehouse operations.
- 🌡️ **Smart sensors** – monitor machine performance and prevent breakdowns.



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3. BIG DATA AND ANALYTICS





- Collects and analyzes **large amounts of data** from machines, IoT devices, ERP, and CRM systems.
- Uses **AI and machine learning** to improve decision-making.
- Helps optimize **production, logistics, maintenance, and inventory management**.

Example:

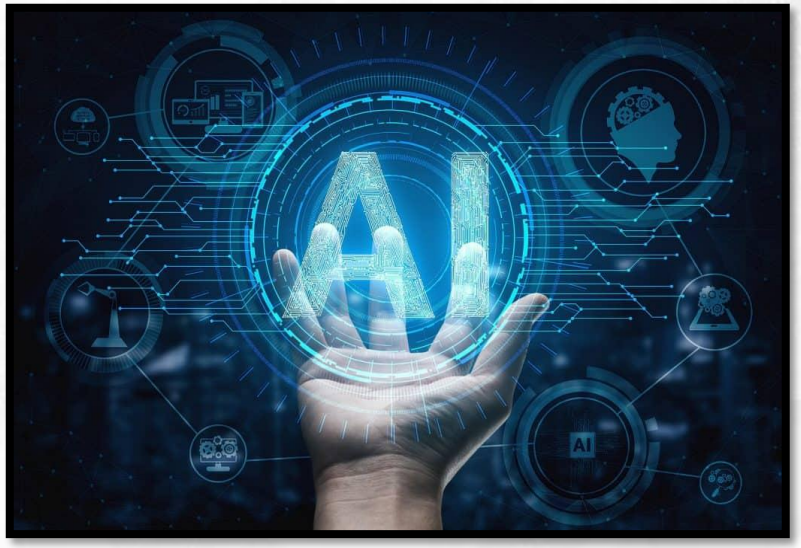
- 🚗 **Tesla** – vehicle performance and production data
- 🎬 **Netflix** – recommendation systems using customer data
- 🥤 **Coca-Cola** – consumer behavior and market analysis
- ✈️ **Delta Air Lines** – predictive maintenance for aircraft
- 💻 **Google** – search, advertising, and user behavior analytics

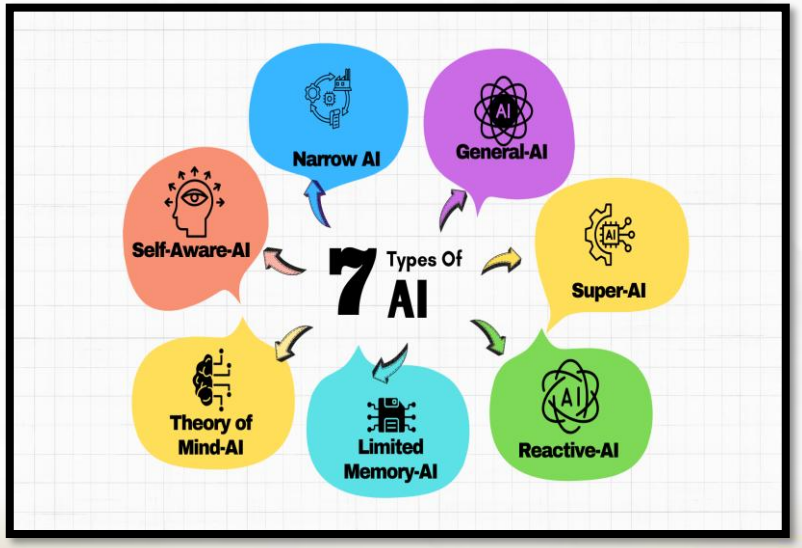
4. ARTIFICIAL INTELLIGENCE (AI)

- Enables machines to **learn, analyze, and make decisions**.
- Uses **machine learning and data** to improve performance.
- Helps automate tasks and increase efficiency.

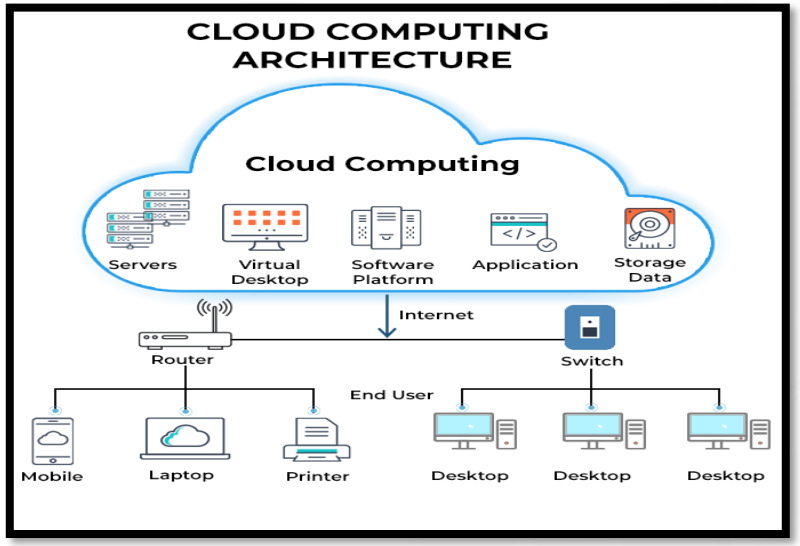
Examples:

- 🎬 **Netflix** – personalized movie and show recommendations.
- 🤖 **ChatGPT (OpenAI)** – AI-powered conversations and content generation.
- 🔍 **Google** – search, translation, and voice assistants.
- 🏭 **Siemens** – predictive maintenance and smart manufacturing.
- 🎤 **Apple Siri** & **Amazon Alexa** – AI-powered virtual assistants.





5. CLOUD COMPUTING





- Stores and processes data over the **internet (cloud)**.
- Enables **real-time access** to information from anywhere.
- Supports **smart factories, IoT devices, and data sharing**.
- Improves **scalability, flexibility, and collaboration**.

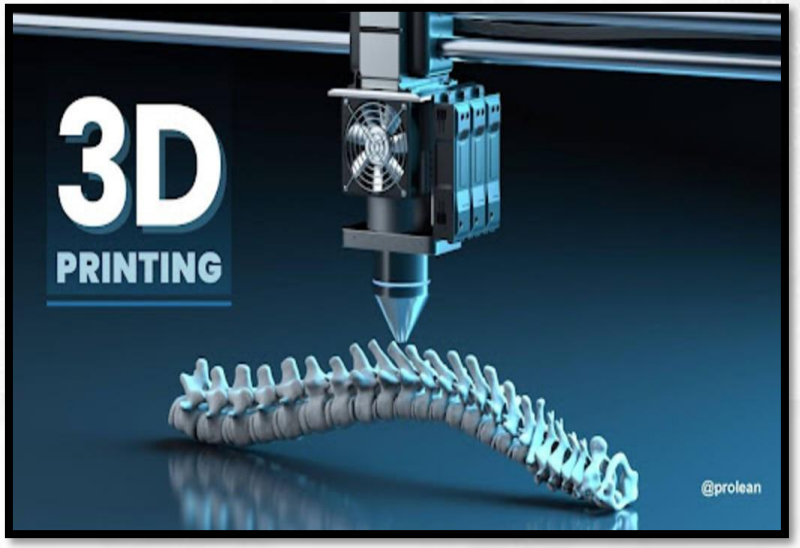
Examples:

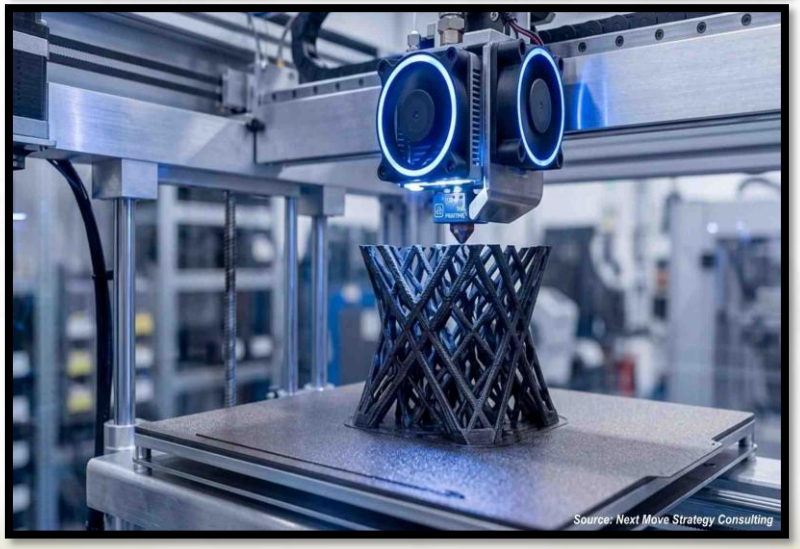
- ☁️ **Microsoft Azure** – cloud platform for smart manufacturing.
- ☁️ **Amazon Web Services (AWS)** – cloud storage and analytics.
- ☁️ **Google Cloud** – AI and cloud-based business solutions.
- 🚗 **Tesla** – stores vehicle data in the cloud.
- 🎬 **Netflix** – delivers streaming services using AWS cloud infrastructure.

6. ADDITIVE MANUFACTURING (3D PRINTING)

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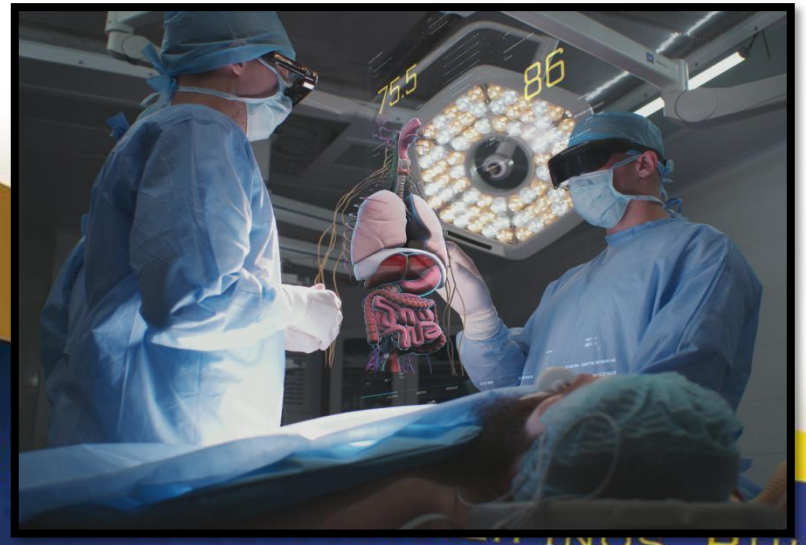
- Creates objects by **adding material layer by layer**.
- Produces **complex parts** with less waste.
- Enables **faster prototyping and customization**.
- Reduces production time and manufacturing costs.
- **Examples:**
- ✈️ **GE Aviation** – 3D-printed aircraft engine parts.
- 🚀 **NASA** – 3D-printed rocket and spacecraft components.
- 🚗 **BMW** – custom car parts and prototypes.
- 🚘 **Ford** – vehicle prototypes and manufacturing tools.
- 🏥 **Johnson & Johnson** – medical devices and implants.
- 👟 **Adidas** – 3D-printed shoe soles.
- 🚀 **SpaceX** – rocket engine parts.





7. AUGMENTED/VIRTUAL REALITY





- **AR** adds digital information to the real world.
- **VR** creates a fully immersive virtual environment.
- Used for **training, maintenance, design, and manufacturing**.
- Improves **productivity, safety, and learning**.

Examples:

- 🚀 **NASA** – VR simulations for astronaut training.
- 🎮 **Meta (Oculus)** – VR headsets and virtual environments.
- 🏥 **Johnson & Johnson** – VR for medical training and surgery practice.
- 🏭 **Bosch** – AR-supported manufacturing and maintenance.
- 🎨 **IKEA** – AR app to visualize furniture in homes before purchase.

8. CYBERSECURITY

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- Protects **networks, systems, and data** from cyber threats.
- Uses **Zero Trust, AI, machine learning, and blockchain** for security.
- Helps detect, prevent, and respond to cyberattacks automatically.
- Reduces the risk of **data breaches and production downtime**.

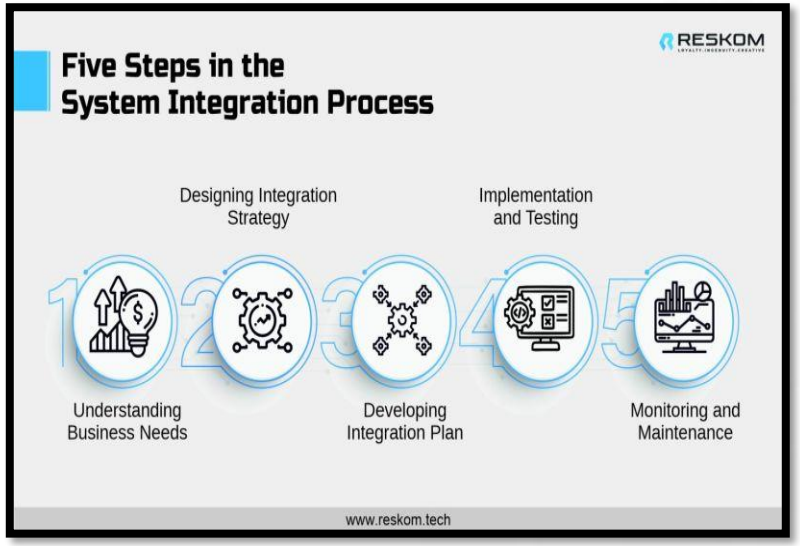
Examples:

- 🔒 **Cisco** – network security for smart factories.
- 💻 **Microsoft Defender** – AI-powered cybersecurity.
- 🔐 **IBM Security** – security monitoring and incident response.
- 🌐 **Fortinet** – industrial and IoT security.
- 🚗 **Tesla** – protects connected vehicles from cyber threats.
- ☁️ **Amazon AWS** – cloud security and data protection.
- ⛓️ **IBM Blockchain** – secure data sharing and transactions.





9. SYSTEM INTEGRATION





- Connects **machines, departments, and business systems** into one network.
- Enables **real-time data sharing and communication**.
- Improves **coordination, efficiency, and decision-making**.
- Links factory operations wit
- h systems like **ERP, CRM, and supply chains**.

Examples:

- ⚙️ **Bosch** – connects smart factories with business operations.
- 🚗 **Toyota** – connects suppliers, factories, and distribution networks.
- 🛒 **Walmart** – links stores, suppliers, and inventory systems.
- ✈️ **Boeing** – integrates design, production, and maintenance systems.
- 💻 **SAP** – provides ERP solutions that connect business processes.
- 🔌 **Schneider Electric** – integrates industrial automation and energy management systems.
- 🚚 **FedEx** – connects logistics, tracking, and delivery operations.

BENEFITS OF INDUSTRY 4.0

- Intelligent Products
- Intelligent Factories
- Intelligent Assets
- Empowered People









BENEFITS OF INDUSTRY 4.0

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Intelligent Products

- Smart products that collect and share data.
- Monitor **location, condition, and usage**.
- Improve product quality and customer service.

Examples:

- 🚗 Tesla connected cars
- ⌚ Apple Watch
- 📦 Smart tracking devices



BENEFITS OF INDUSTRY 4.0

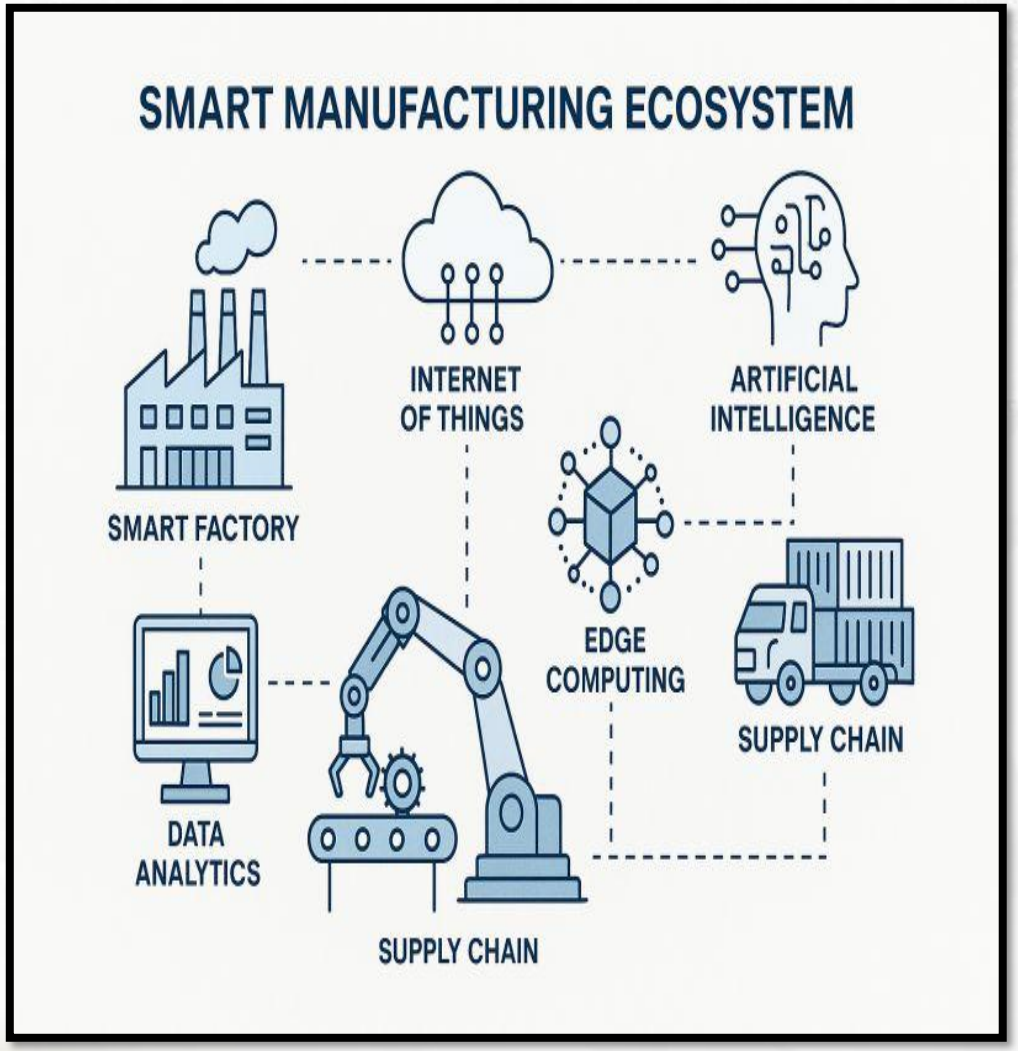
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Intelligent Factories (Factory 4.0)

- Highly automated and digital factories.
- Use **AI, IoT, robotics, and Big Data**.
- Produce customized products efficiently.

Examples:

- 🏭 Siemens Amberg Factory
- 🚗 Tesla Gigafactory
- ⚙️ Bosch Smart Factory



BENEFITS OF INDUSTRY 4.0

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Intelligent Assets

- Machines with sensors that provide real-time data.
- Enable **predictive maintenance** and reduce downtime.
- Improve asset performance and monitoring.

Examples:

- ✈️ GE aircraft engines
- 🚂 Siemens trains
- 🏭 Caterpillar smart equipment



BENEFITS OF INDUSTRY 4.0

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Empowered People

- Workers use **AI, wearable devices, and AR/VR** tools.
- Improve productivity, safety, and decision-making.
- Access real-time information.

Examples:

- 🥽 Microsoft HoloLens
- ✈️ Boeing AR glasses
- ⌚ Smart wearable safety devices
- 🚗 BMW VR training systems



COMPANIES ARE EXPERIENCING THE BENEFITS

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- Radical improvements in productivity and automation
- Resiliency and agility no matter what the market or economy bring
- Confidence to explore new business models and seize opportunities quickly
- Green and sustainable solutions without sacrificing profitability



INDUSTRY 4.0: OPPORTUNITIES AND CHALLENGES

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What Businesses Need to Do

- Upgrade **internet and technology infrastructure** to be fast, secure, and reliable.
- Improve **education and training programs** to develop IT and digital skills.
- Encourage **innovation and entrepreneurship** among workers and students.
- Foster cooperation between **government, industries, and businesses**.

Opportunities

- Better productivity and efficiency.
- Increased innovation and competitiveness.
- Development of smart factories and digital businesses.
- Creation of new products and services.

Challenges

- High cost of technology adoption.
- Need for advanced digital skills.
- Risk for companies that fail to adapt.
- Changes in job requirements and business models.

CONCLUSION

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- **Industry 4.0** is transforming industries through **automation, connectivity, and intelligent technologies**.
- It is built on technologies such as **AI, IoT, Big Data, Cloud Computing, Robotics, Cybersecurity, System Integration, Additive Manufacturing, and AR/VR**.
- These innovations create **smart products, smart factories, intelligent assets, and empowered workers**.
- Businesses can achieve **higher productivity, efficiency, quality, and competitiveness**.
- Success in Industry 4.0 requires **strong digital infrastructure, skilled workers, and continuous innovation**.
- Companies that embrace Industry 4.0 will be better prepared for the future, while those that fail to adapt may fall behind.

ACTIVITY

Industry 4.0 Technologies

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Objective: Identify the different Industry 4.0 technologies and their real-world applications.

Instructions:

- Form groups of 4-5 students.
- Choose any 3 Industry 4.0 technologies together with their company.
- Explain in 3-5 sentences how the company uses the technology.

Guide Questions

- What Industry 4.0 technology is the most important today? Why?
- How do these technologies improve business operations?
- What challenges might companies face when adopting Industry 4.0?

Industry 4.0 Technology	Company
1. Advanced Tobotics	Coca-cola Beverages Phil.
2. Industrial IoT	Meralco
3. Big Data & Analytics	SM Supermalls
4. AI	GCash
5. Cloud Computing	PLDT
6. 3D Printing	DOST
7. AR/VR	Ayala Land
8. Cybersecurity	BDO
9. System Integration	San Miguel

REFERENCES:

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- https://www.moderncasting.com/articles/2021/04/20/practical-implementation-industry-40-foundries
- https://www.ibm.com/topics/industry-4-0
- https://insights.sap.com/what-is-industry-4-0/
- https://www.britannica.com/topic/The-Fourth-Industrial-Revolution-2119734
- https://www.avnet.com/wps/portal/abacus/resources/article/the-origins-of-industry-4-0/
- https://www.iotworldtoday.com/2016/11/02/short-history-fourth-industrial-revolution/

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Thank you!

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